

PHYS 217 Electricity and Magnetism I

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Contents

1	Vector Analysis	2
1.1	Vector Algebra	2
1.1.1	Vector Operations	2
1.1.2	Nomenclature	3
1.1.3	Vector Transformations	3
1.1.4	Dyadics and Tensors	3
1.2	Differential Calculus	4
1.3	Integral Calculus	5
1.4	Curvilinear Coordinates	6
1.5	Dirac Delta Function	7
1.6	Theory of Vector Fields	7
2	Electrostatics	8
2.1	The Electric Field	8
2.2	Divergence and Curl of Electrostatic Fields	8
2.3	Electric Potential	8
2.4	Work and Energy in Electrostatics	9

1 Vector Analysis

1.1 Vector Algebra

Definition 1.1. A **vector** is a quantity that has both direction and magnitude.

Definition 1.2. A **scalar** is a quantity that only has magnitude.

1.1.1 Vector Operations

There are four vector operations:

Addition:

- $\mathbf{A} + \mathbf{B}$ is the vector from the start of $\mathbf{A}$ to the end of $\mathbf{B}$
- Commutative: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$
- Associative: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$
- Subtracting a vector is just adding its opposite: $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$

Multiplication by Scalar:

- $a\mathbf{A}$ is a vector with the same direction as $\mathbf{A}$ but a times the magnitude
- Distributive: $a(\mathbf{A} + \mathbf{B}) = a\mathbf{A} + a\mathbf{B}$

Dot Product (Inner Product or Scalar Product):

- $\mathbf{A} \cdot \mathbf{B} \equiv AB \cos \theta$
- Results in scalar
- Commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- Distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$
- Geometrically, $\mathbf{A} \cdot \mathbf{B}$ is the product of A and the projection of $\mathbf{A}$ along $\mathbf{B}$

Cross Product of Two Vectors:

- $\mathbf{A} \times \mathbf{B} \equiv AB \sin \theta \hat{\mathbf{n}}$ where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the plane of $\mathbf{A}$ and $\mathbf{B}$. The direction is determined by the **right hand rule**, where you curl your right hand from $\mathbf{A}$ to $\mathbf{B}$ and take the direction of your thumb
- Distributive: $\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$
- Not commutative: $\mathbf{B} \times \mathbf{A} = -(\mathbf{A} \times \mathbf{B})$
- Result is a **pseudovector** and behaves like a vector under rigid transformations like rotations, but not under certain nonrigid transformations. Notably, $-\mathbf{A} \times -\mathbf{B} = \mathbf{A} \times \mathbf{B}$
- Geometrically, $|\mathbf{A} \times \mathbf{B}|$ is the area of the parallelogram with sides $\mathbf{A}$ and $\mathbf{B}$

When working in Cartesian coordinates, vectors can be expressed in terms of **basis vectors**:

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

with A_x , A_y , and A_z being the **components** of $\mathbf{A}$.

Vector operations can be done with the components, but the reader should already know how to do this.

There are two types of triple vector products:

Scalar Triple Product:

- $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})$
- Volume of parallelepiped generated by $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$

Vector Triple Product:

- **BAC-CAB rule:** $\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$

1.1.2 Nomenclature

Definition 1.3. The **position vector** of a point is the vector to that point from the origin:

$$\mathbf{r} \equiv x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

Definition 1.4. The **infinitesimal displacement vector** from (x, y, z) to $(x + dx, y + dy, z + dz)$ is:

$$d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$$

Definition 1.5. A **separation vector** is the vector from a **source point** to a **field point**:

$$\mathbf{r} \equiv \mathbf{r} - \mathbf{r}'$$

1.1.3 Vector Transformations

Coordination transformation from rotation about an arbitrary axis can be represented by the use of a **rotation matrix**:

$$\begin{pmatrix} \bar{A}_x \\ \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{yx} & R_{yy} & R_{yz} \\ R_{zx} & R_{zy} & R_{zz} \end{pmatrix} \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix}$$

or more concisely:

$$\bar{A}_i = \sum_{j=1}^3 R_{ij} A_j$$

Definition 1.6. A transformation is **unitary** if it does not change the magnitude of the object on which it operates.

If rotation matrix $\overleftrightarrow{R}$ is unitary, then:

$$\sum_i R_{ij} R_{ik} = \delta_{jk}$$

where:

$$\delta_{jk} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

is the Kronecker delta.

Definition 1.7. **Parity** is a property related to how a system is affected by coordinate inversions. If a system stays the same after a parity transformation, it is said to have even parity. If it is inverted, it is said to have odd parity.

1.1.4 Dyadics and Tensors

Definition 1.8. A **tensor** is something that transforms like a tensor.

Definition 1.9. A **dyadic product** or outer product is a type of multiplication that generates a second-order tensor called a **dyadic** from two vectors:

$$\overleftrightarrow{\mathbf{C}} = \mathbf{A}\mathbf{B}$$

Dyadics have nine independent components and in Cartesian coordinates can be decomposed into **dyads**:

$$\overleftrightarrow{\mathbf{C}} = A_x B_x \hat{\mathbf{x}}^2 + A_x B_y \hat{\mathbf{x}}\hat{\mathbf{y}} + A_x B_z \hat{\mathbf{x}}\hat{\mathbf{z}} + A_y B_x \hat{\mathbf{y}}\hat{\mathbf{x}} + A_y B_y \hat{\mathbf{y}}^2 + A_y B_z \hat{\mathbf{y}}\hat{\mathbf{z}} + A_z B_x \hat{\mathbf{z}}\hat{\mathbf{x}} + A_z B_y \hat{\mathbf{z}}\hat{\mathbf{y}} + A_z B_z \hat{\mathbf{z}}^2$$

Thus, dyadics can be represented in matrix form:

$$\overleftrightarrow{\mathbf{C}} = \begin{bmatrix} A_x B_x & A_x B_y & A_x B_z \\ A_y B_x & A_y B_y & A_y B_z \\ A_z B_x & A_z B_y & A_z B_z \end{bmatrix} = \begin{bmatrix} A_x & A_y & A_z \end{bmatrix} \begin{bmatrix} B_x \\ B_y \\ B_z \end{bmatrix} = \mathbf{A} \mathbf{B}^T$$

The transpose symbol is usually omitted in E&M.

A dyadic $\overleftrightarrow{\mathbf{C}}$ has two invariants, **spur** (called cusp in lecture) $|\overleftrightarrow{\mathbf{C}}|$ and **rotation vector** $\langle \overleftrightarrow{\mathbf{C}} \rangle$:

$$|\overleftrightarrow{\mathbf{C}}| = \sum_i^3 C_{ii}$$

$$\langle \overleftrightarrow{\mathbf{C}} \rangle = \hat{\mathbf{x}}(C_{yz} - C_{zy}) + \hat{\mathbf{y}}(C_{zx} - C_{xz}) + \hat{\mathbf{z}}(C_{xy} - C_{yx})$$

Dyadics commute in addition and scalar multiplication but not in inner products:

$$\begin{aligned} \overleftrightarrow{\mathbf{C}} + \overleftrightarrow{\mathbf{D}} &= \overleftrightarrow{\mathbf{D}} + \overleftrightarrow{\mathbf{C}} \\ a \overleftrightarrow{\mathbf{C}} &= \overleftrightarrow{\mathbf{C}} a \\ \mathbf{A} \cdot \overleftrightarrow{\mathbf{C}} &\neq \overleftrightarrow{\mathbf{C}} \cdot \mathbf{A} \end{aligned}$$

Definition 1.10. Every dyadic $\overleftrightarrow{\mathbf{C}}$ has a **conjugate** $\overleftrightarrow{\mathbf{C}}^*$ such that:

$$\mathbf{A} \cdot \overleftrightarrow{\mathbf{C}} = \overleftrightarrow{\mathbf{C}}^* \cdot \mathbf{A}$$

In three dimensional Cartesian coordinates, the conjugate of a dyadic is simply its transpose.

1.2 Differential Calculus

Definition 1.11. The **gradient** of a three dimensional scalar-valued function $f(x, y, z)$ is a vector field that points in the direction of maximum increase of f and has a magnitude equal to the rate of increase:

$$\nabla f = \frac{\partial f}{\partial x} \hat{\mathbf{d}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$$

Definition 1.12. The **divergence** of a three dimensional vector-valued function $f(x, y, z) = f_x \hat{\mathbf{x}} + f_y \hat{\mathbf{y}} + f_z \hat{\mathbf{z}}$ is a scalar that represents how much the vector field diverges or converges, analogous to sources and sinks in fluid flow:

$$\nabla \cdot f = \frac{\partial f_x}{\partial x} + \frac{\partial f_y}{\partial y} + \frac{\partial f_z}{\partial z}$$

Definition 1.13. The **curl** of a three dimensional vector-valued function $f(x, y, z) = f_x \hat{\mathbf{x}} + f_y \hat{\mathbf{y}} + f_z \hat{\mathbf{z}}$ is a vector field with magnitude representing the infinitesimal circulation, analogous to rotation in a fluid, and direction perpendicular to that circulation:

$$\nabla \times f = \left(\frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial f_x}{\partial z} - \frac{\partial f_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} \right) \hat{\mathbf{z}}$$

Gradient, divergence, and curl can all be found by treating the vector operator ∇ **del** as a vector:

$$\nabla = \hat{\mathbf{x}} \frac{\partial}{\partial x} + \hat{\mathbf{y}} \frac{\partial}{\partial y} + \hat{\mathbf{z}} \frac{\partial}{\partial z}$$

Gradient, divergence, and curl behave like ordinary derivatives under summation and multiplication by scalars. Since gradient, divergence, and curl can be multiplied by scalars and vectors in multiple

ways, there are six product rules:

$$\begin{aligned}
 \nabla(fg) &= f\nabla g + g\nabla f \\
 \nabla(\mathbf{A} \cdot \mathbf{B}) &= \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \\
 \nabla \cdot (f\mathbf{A}) &= f(\nabla \cdot \mathbf{A}) + \mathbf{A} \cdot (\nabla f) \\
 \nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\
 \nabla \times (f\mathbf{A}) &= f(\nabla \times \mathbf{A}) - \mathbf{A} \times (\nabla f) \\
 \nabla \times (\mathbf{A} \times \mathbf{B}) &= (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B} + \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A})
 \end{aligned}$$

Since you can combine the three derivatives in various ways, there are five species of second derivatives, but all but two are either degenerate or are composed of the others:

$$\begin{aligned}
 \nabla \cdot (\nabla f) \\
 \nabla(\nabla \cdot f)
 \end{aligned}$$

Definition 1.14. The **Laplacian** of three dimensional scalar-valued function f is:

$$\nabla^2 f = \nabla \cdot (\nabla f)$$

1.3 Integral Calculus

Definition 1.15. A **line integral** of a vector-valued function is the integral of the dot product of the function and infinitesimal displacement along a curve:

$$\int_a^b \mathbf{v} \cdot d\mathbf{l}$$

Definition 1.16. A **surface integral** of a vector-valued function is the integral of the dot product of the function and infinitesimal area over a surface:

$$\int_S \mathbf{v} \cdot d\mathbf{a}$$

Definition 1.17. A **volume integral** of a scalar-valued function is the integral of the function over a volume:

$$\int_v T d\tau$$

Since there are three types of derivatives in three dimensions, there are three versions of the fundamental theorem of calculus.

Theorem 1.1. Gradient Theorem

$$\int_a^b (\nabla T) \cdot d\mathbf{l} = T(\mathbf{b}) - T(\mathbf{a})$$

Proof. left as an exercise for the reader (will do tonight) □

Theorem 1.2. Divergence Theorem or Gauss's Theorem

$$\int_V (\nabla \cdot \mathbf{v}) d\tau = \oint_{\partial V} \mathbf{v} \cdot d\mathbf{a}$$

Proof. left as an exercise for the reader (will finish tonight) □

Theorem 1.3. Stoke's Theorem

$$\int_S (\nabla \times \mathbf{v}) d\mathbf{a} = \oint_{\partial S} \mathbf{v} \cdot d\mathbf{l}$$

Proof. left as an exercise for the reader (will finish tonight) □

1.4 Curvilinear Coordinates

Definition 1.18. Spherical coordinates are a coordinate system that uses the distance from the origin r , the angle down from the z-axis called the **polar angle** θ , and the angle around from the x-axis called the **azimuthal angle** ϕ .

The following are true for spherical coordinates:

- Conversion to Cartesian coordinates: $x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$
- Unit vectors in terms of Cartesian coordinates:

$$\begin{aligned}\hat{\mathbf{r}} &= \sin \theta \cos \phi \hat{\mathbf{x}} + \sin \theta \sin \phi \hat{\mathbf{y}} + \cos \theta \hat{\mathbf{z}} \\ \hat{\theta} &= \cos \theta \cos \phi \hat{\mathbf{x}} + \cos \theta \sin \phi \hat{\mathbf{y}} - \sin \theta \hat{\mathbf{z}} \\ \hat{\phi} &= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}}\end{aligned}$$

- Infinitesimal displacement: $d\mathbf{l} = dr\hat{\mathbf{r}} + r d\theta\hat{\theta} + r \sin \theta d\phi\hat{\phi}$
- Infinitesimal volume: $d\tau = r^2 \sin \theta dr d\theta d\phi$
- Gradient: $\nabla T = \frac{\partial T}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\phi}$
- Divergence: $\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$
- Curl: $\nabla \times \mathbf{v} = \frac{1}{r \sin \theta} [\frac{\partial}{\partial \theta} (\sin \theta v_\phi) - \frac{\partial v_\theta}{\partial \phi}] \hat{\mathbf{r}} + \frac{1}{r} [\frac{1}{\sin \theta} \frac{\partial v_r}{\partial \phi} - \frac{\partial}{\partial r} (r v_\phi)] \hat{\theta} + \frac{1}{r} [\frac{\partial}{\partial r} (r v_\theta) - \frac{\partial v_r}{\partial \theta}] \hat{\phi}$
- Laplacian: $\nabla^2 T = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial T}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial T}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2}$

Definition 1.19. Cylindrical coordinates are a coordinate system that uses distance from the z-axis s , the angle around from the x-axis ϕ , and the distance up the z-axis z .

The following are true for cylindrical coordinates:

- Conversion to Cartesian coordinates: $x = s \cos \phi$, $y = s \sin \phi$, $z = z$
- Unit vectors in terms of Cartesian coordinates:

$$\begin{aligned}\hat{\mathbf{s}} &= \cos \phi \hat{\mathbf{x}} + \sin \phi \hat{\mathbf{y}} \\ \hat{\phi} &= -\sin \phi \hat{\mathbf{x}} + \cos \phi \hat{\mathbf{y}} \\ \hat{\mathbf{z}} &= \hat{\mathbf{z}}\end{aligned}$$

- Infinitesimal displacement: $d\mathbf{l} = ds\hat{\mathbf{s}} + s d\phi\hat{\phi} + dz\hat{\mathbf{z}}$
- Infinitesimal volume: $d\tau = s ds d\phi dz$
- Gradient: $\nabla T = \frac{\partial T}{\partial s} \hat{\mathbf{s}} + \frac{1}{s} \frac{\partial T}{\partial \phi} \hat{\phi} + \frac{\partial T}{\partial z} \hat{\mathbf{z}}$
- Divergence: $\nabla \cdot \mathbf{v} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$
- Curl: $\nabla \times \mathbf{v} = (\frac{1}{s} \frac{\partial v_z}{\partial \phi} - \frac{\partial v_\phi}{\partial z}) \hat{\mathbf{s}} + (\frac{\partial v_s}{\partial z} - \frac{\partial v_z}{\partial s}) \hat{\phi} + \frac{1}{s} [\frac{\partial}{\partial s} (s v_\phi) - \frac{\partial v_s}{\partial \phi}] \hat{\mathbf{z}}$
- Laplacian: $\nabla^2 T = \frac{1}{s} \frac{\partial}{\partial s} (s \frac{\partial T}{\partial s}) + \frac{1}{s^2} \frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2}$

1.5 Dirac Delta Function

Definition 1.20. The **Dirac delta function** is a **generalized function** or **distribution** with a value of 0 everywhere except 0, where it equals infinity:

$$\delta(x) = \begin{cases} 0 & x \neq 0 \\ \infty & x = 0 \end{cases}$$

The delta function has the integral:

$$\int_{-\infty}^{\infty} \delta(x) dx = 1$$

The Dirac delta function is useful because it allows us to pick out a specific value of a function in an integral:

$$\int_{-\infty}^{\infty} f(x) \delta(x - a) dx = f(a)$$

The Dirac delta function behaves similarly in three dimensions and can be written as:

$$\delta^3(\mathbf{r}) = \frac{1}{4\pi} \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right)$$

1.6 Theory of Vector Fields

Theorem 1.4. Helmholtz Theorem

If the divergence $D(\mathbf{r})$ and curl $C(\mathbf{r})$ of a vector function $\mathbf{F}(\mathbf{r})$ are specified, and if both go to zero faster than $\frac{1}{r^2}$ as $r \rightarrow \infty$, and if $\mathbf{F}(\mathbf{r})$ goes to zero as $r \rightarrow \infty$, then $\mathbf{F}$ can uniquely be decomposed into an irrotational component $-\nabla U$ and solenoidal component $\nabla \times \mathbf{W}$:

$$\mathbf{F} = -\nabla U + \nabla \times \mathbf{W}$$

Theorem 1.5. Theorem 1. Irrotational Fields

The following conditions are equivalent:

- $\nabla \times \mathbf{F} = 0$ everywhere
- $\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{l}$ is independent of path for any given end points
- $\oint \mathbf{F} \cdot d\mathbf{l} = 0$ for any closed loop
- $\mathbf{F}$ is the gradient of some scalar function V called the **scalar potential**: $\mathbf{F} = -\nabla V$

Theorem 1.6. Theorem 2. Solenoidal Fields

The following conditions are equivalent:

- $\nabla \cdot \mathbf{F} = 0$ everywhere
- $\int \mathbf{F} \cdot d\mathbf{a}$ is independent of surface for a given boundary line
- $\oint \mathbf{F} \cdot d\mathbf{a} = 0$ for any closed surface
- $\mathbf{F}$ is the curl of some vector function $\mathbf{A}$ called the **vector potential**: $\mathbf{F} = \nabla \times \mathbf{A}$

2 Electrostatics

2.1 The Electric Field

Definition 2.1. The **principle of superposition** states that the interaction between any two charges is completely unaffected by the presence of other charges.

Law 2.1. Coulomb's Law

The force on charge Q from charge q with separation vector $\mathbf{z}$ between them is:

$$\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qQ}{z^2} \hat{\mathbf{z}}$$

where ϵ_0 is the **permittivity of free space**.

Definition 2.2. An **electric field** $\mathbf{E}$ is the total contribution from a collection of charges or a continuous charge distribution, with the resultant force on test charge Q being:

$$\mathbf{F} = Q\mathbf{E}$$

The electric field of a volume charge with charge density ρ is:

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}')}{z^2} \hat{\mathbf{z}} d\tau'$$

2.2 Divergence and Curl of Electrostatic Fields

We notice that the **electric flux** $\int \mathbf{E} \cdot d\mathbf{a}$ through a closed surface only depends on the divergence of the electric field within that surface, since any fields originating from outside pass through two sides of the surface, making the net flux zero.

Law 2.2. Gauss's Law

Integral form:

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{Q_{enc}}{\epsilon_0}$$

Differential form:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

We can also derive this by taking the divergence of the formula for the electric field from a charge distribution:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{4\pi\epsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{z}}}{z^2} \right) \rho(\mathbf{r}') d\tau' \\ &= \frac{1}{4\pi\epsilon_0} \int 4\pi\delta^3(\mathbf{r} - \mathbf{r}') \rho(\mathbf{r}') d\tau' \\ &= \frac{\rho(\mathbf{r})}{\epsilon_0} \end{aligned}$$

When a situation has symmetry, Gauss's law can be used to easily find the electric field. Additionally, due to the principle of superposition, some situations can be split into multiple problems that have symmetry and are thus simple to solve.

2.3 Electric Potential

Definition 2.3. The **electric potential** is defined as:

$$V(\mathbf{r}) = \int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\mathbf{l}$$

where $\mathcal{O}$ is a reference point where the potential is 0 (typically at infinity). This can be thought of as the work required to move a unit charge from infinity to $\mathbf{r}$.

The differential form of this is:

$$\mathbf{E} = -\nabla V$$

Electric potential is useful because it is a scalar function, not a vector function (though the electric field still only has one independent function because it has zero curl). Like the electric field, it obeys the principle of superposition.

Definition 2.4. Poisson's Equation for Electrostatics

$$\nabla^2 V = -\frac{\rho}{\epsilon_0}$$

Definition 2.5. Laplace's Equation

$$\nabla^2 V = 0$$

The potential of a charge distribution is:

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{r} d\tau'$$

The electric field experiences discontinuities across surface charges (easily verified using Gauss's law) but the tangential component is continuous across surface charges (also easily verified with Gauss's law), so:

$$E_{above}^\perp - E_{below}^\perp = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}}$$

This can also be written in terms of the electric potential:

$$\frac{\partial V_{above}}{\partial n} - \frac{\partial V_{below}}{\partial n} = -\frac{\sigma}{\epsilon_0}$$

2.4 Work and Energy in Electrostatics

Starting with a single point charge, we can incrementally move point charges from infinity to different positions, with the total amount of work done being the energy of the configuration:

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i)$$

We can determine a version of this for continuous charge distributions:

$$\begin{aligned} W &= \frac{1}{2} \int \rho V d\tau \\ &= \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau \quad \text{using Gauss's law} \\ &= \frac{\epsilon_0}{2} \left[- \int_{\mathcal{V}} \mathbf{E} \cdot (\nabla V) d\tau + \oint_{\mathcal{S}} V \mathbf{E} \cdot d\mathbf{a} \right] \quad \text{using integration by parts} \\ &= \frac{\epsilon_0}{2} \left(\int_{\mathcal{V}} E^2 d\tau + \oint_{\mathcal{S}} V \mathbf{E} \cdot d\mathbf{a} \right) \\ &= \frac{\epsilon_0}{2} \int E^2 d\tau \quad \text{if } \mathbf{E} \text{ vanishes at infinity} \end{aligned}$$